

Matthew Whispell

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Electrical Engineer with a focus on EV design, with experience in low and high voltage circuitry and design for electrification. Additionally, I enjoy control theory and embedded systems.

Professional Experience

Clarkson Formula SAE Electric Racing Team • Electrical Engineering 2025-Present

- Created modular dash software for ESP32-S3 that interfaces with CANbus to provide the driver a data rich interface that also allows for live configuration parameters to be tuned.
- Utilized Electrical Safety Officer training to ensure that procedures for handling high voltage systems are followed.
- Developed a battery pack that provides 25% more current and 30% more voltage than the last design to provide higher torque and speed output.

Clarkson University Undergraduate Research • Embedded Systems May-July '26

- Firmware development of a custom environmental monitoring device built on the Silicon Labs EFR32 SoC and TI LMP91000 AFE.
- Added functionality to detect pesticides using electrochemical techniques, pioneering edge detection of environmental contaminants.

Hazen Enterprises • Maxfields and Cactus Grill & Cantina Restaurants 2021-Present

- Interfaced between the kitchen and front-of-house to ensure food was sent correctly.
- Worked on a team of food runners to keep kitchen output consistent.

Potsdam FTC Robotics Club • Team Lead / Lead Engineer 2020-2024

- Delegated work to sub-teams, leading to higher productivity & more efficient use of time.
- Trained machine-learning models for use in autonomous navigation and object manipulation, leading to 20% higher average match scores.

Education

Clarkson University • Bachelor of Electrical Engineering • 3.98 Cumulative GPA 2024-2028

- Awards: Presidential Scholar '24-'26, Undergraduate Commercialization Award 2025

Potsdam Senior High School • Academic Top 5 • 100.39 Cumulative Average 2020-2024

- Awards: FTC Innovation Award, FTC Robotics State Championship winner (x2)

Skills

High Voltage Safety, Altium Designer, Embedded Systems, C++, C#, Python, CANbus Protocol, Verilog HDL, MATLAB, TensorFlow Pipelines, Machine Learning Training, PID controllers, 3D Printing, Autodesk Fusion, Systems Integration, Team Leadership

Projects

High Voltage Battery Pack Design, EV Drivetrain Electrical Design, Custom EEPROM Programmer, Adjustable Clock Circuit, Stepper Driver Circuit Design, Custom 3D Printer Design